

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

ASSESSMENT TOOL 1 – CONCEPT MAPPING

Course Code	CSA0314	Course Title	Data Structures
CO Assessed	CO1	Assessment	Assessment Tool 1: Concept Mapping
Weightage	35% of CIA	Total Marks	100
CO4 Statement	<i>Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL2, BL3)</i>		
Student Name	Dinakaran.S	Reg. No.	1913260107
Section/Batch	Slot C / 2026	Date	06-10-2026

Group 2 **Code of Conduct**

I, Dinakaran.S, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

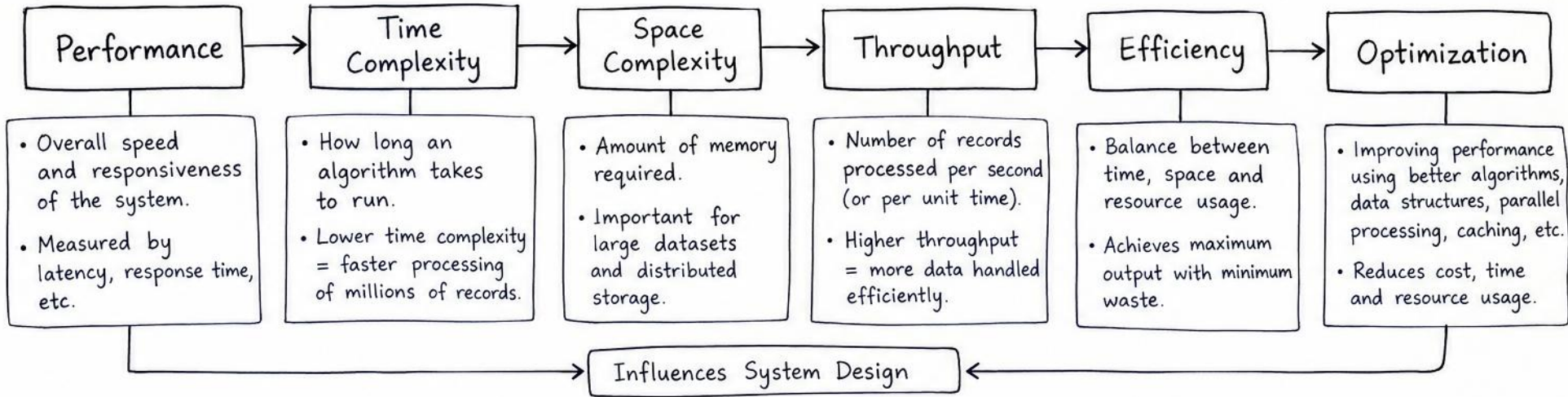


Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts	
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established	
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure	
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence	
Clarity	10%	Concept map is clear, neat,	Generally clear with minor	Clarity is reduced;	Poorly presented and hard to	

		visually structured, and easy to interpret	readability issues	difficult to interpret in parts	understand	
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Concept Map - Big Data System

(Performance Metrics and Optimization)



How Performance Metrics Influence System Design

- Time complexity → guides algorithm selection and data partitioning.
- Space complexity → decides memory allocation and storage model.
- Throughput → influences the number of nodes, parallelism and hardware.
- Efficiency → affects resource management, load balancing and fault tolerance.
- Overall performance → shapes the architecture (e.g., batch vs real-time processing, distributed systems, caching, etc.).

Justification for Optimization Strategies

- Use efficient algorithms and data structures (e.g., B-trees, hash tables) → reduces time complexity.
- Minimize memory usage (e.g., compression, data deduplication) → lowers space complexity.
- Scale horizontally (add more nodes) → increases throughput.
- Implement caching and indexing → improves speed and efficiency.
- Use parallel and distributed processing (e.g., MapReduce, Spark) → handles millions of records effectively.
- Monitor and adjust (auto-scaling, load balancing) → maintains high performance and efficiency.

Result

By balancing time, space, throughput and efficiency, and applying appropriate optimization strategies, the big data system can process millions of records faster, with lower cost and better reliability.